Universidad Nacional Abierta y a Distancia

Vicerrectoría Académica y de Investigación

Unidad gestora: *ECBTI*

Curso: Estructuras de Datos

Código: 301305A_2204

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INTRODUCTION

In this practical work, we developed a desktop application using Python and the Tkinter library to apply linear data structures like stacks, queues, and lists. The main goal was to build a system for a family compensation fund called "Compensándote" to manage user registrations, calculate fees based on income and services, and handle data storage correctly using programming fundamentals learned in the course.

OBJETIVES

General Objective:

To design and program a Python application with a graphical user interface that uses linear data structures (stacks, queues, and lists) to solve an affiliation management problem.

Specific Objectives:

- To create a functional login screen and a registration form following the requirements of the case study.

- To implement object-oriented programming concepts to organize affiliate information and compute fees.

- To test how stack (LIFO) and queue (FIFO) data structures work in a real software scenario.

TABLA 1. ABSTRACCIÓN DE DATOS PARA EL SISTEMA DE AFILIACIONES.

Nombre de la Clase y Ámbito de Visibilidad	Nombre de las Propiedades y/o Atributos con sus Tipos de Datos	Tipo de Estructura que Soporta (Pila, Cola o Lista)	Acciones que tendrá el Formulario de Ingreso (Métodos)
Public Class Affiliate	id_type: String id_number: String full_name: String income: float service: String employment_type: String affiliation_fee: float date: String	Soporta estructuras lineales: - **Pila** (Stack) - **Cola** (Queue) - **Lista** (List)	- registrar() - limpiar() - reporte() - eliminar()

LINK:

https://drive.google.com/file/d/1q5opknCyvajaZwiPx-2Cw4jh9ow68jXG/view?usp=sharing

CONCLUSIONS

Working on this practical activity helped us understand how linear data structures work in real Python code. Using stacks and queues made it clear how data is stored and retrieved differently depending on whether it is LIFO or FIFO. Also, combining these concepts with a Tkinter graphical interface allowed us to build a complete and functional program, which is very useful for future software development projects.

REFERENCES

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